

ATS-MINI DISPLAY

I had a quick look at the datasheets for the ST7789 and GC9307 display drivers, and the bitmaps essentially appear to be compatible.

The overall behaviour depends on the actual display as well.

Example: Setup206_LilyGo_T_Display_S3.h

```
#define TFT_INVERSION_ON
// #define TFT_INVERSION_OFF
```

It sort of seemed odd to me that TFT_INVERSION is set to "ON", but it works correctly for the LilyGo_T_Display_S3 display.

For the SI4732 receiver with the different display, the setting would need to be TFT_INVERSION_OFF or it can be set by tft.invertDisplay(0);

The same is likely to be true for the MADCTL register bits, which controls the Rotation and RGB ordering. The TFT_eSPI library controls this with various rotation options in ST7789_Rotation.h file.

9.1.28 MADCTL (36h): Memory Data Access Control

36H	MADCTL (Memory Data Access Control)																																	
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX																					
MADCTL	0	↑	1	-	0	0	1	1	0	1	1	0	(36h)																					
parameter	1	↑	1	-	MY	MX	MV	ML	RGB	MH	-	-																						
-This command defines read/ write scanning direction of frame memory.																																		
<table><tr><th>Bit</th><th>NAME</th><th>DESCRIPTION</th></tr><tr><td>D7</td><td>MY</td><td>Page Address Order</td></tr><tr><td>D6</td><td>MX</td><td>Column Address Order</td></tr><tr><td>D5</td><td>MV</td><td>Page/Column Order</td></tr><tr><td>D4</td><td>ML</td><td>Line Address Order</td></tr><tr><td>D3</td><td>RGB</td><td>RGB/BGR Order</td></tr><tr><td>D2</td><td>MH</td><td>Display Data Latch Order</td></tr></table>														Bit	NAME	DESCRIPTION	D7	MY	Page Address Order	D6	MX	Column Address Order	D5	MV	Page/Column Order	D4	ML	Line Address Order	D3	RGB	RGB/BGR Order	D2	MH	Display Data Latch Order
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The setting for inversion and rotation can be controlled in the user code.

These are the settings that work with the receiver that I have.

```
// TFT display setup
tft.begin();
tft.invertDisplay(1);
tft.setRotation(3);
```

The current options are:

```
tft.invertDisplay(*); * = 0, 1
tft.setRotation(*); * = 0, 1, 2, 3
```

The results are shown below for the receiver that I have.

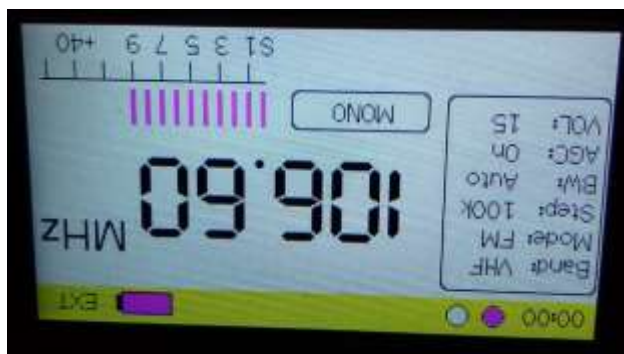
```
tft.invertDisplay(1);  
tft.setRotation(3);
```



```
tft.invertDisplay(0);  
tft.setRotation(3);
```



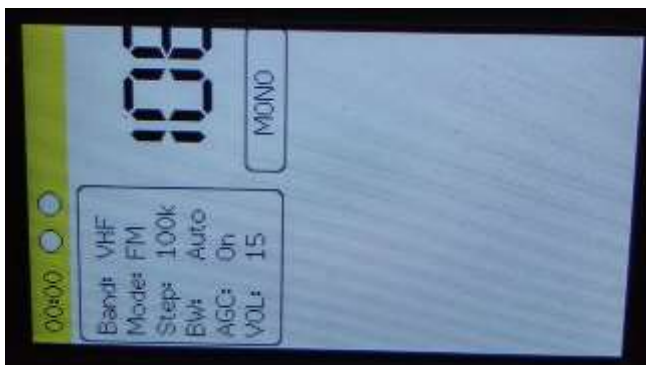
```
tft.invertDisplay(0);  
tft.setRotation(1);
```



```
tft.invertDisplay(0);  
tft.setRotation(0);
```



```
tft.invertDisplay(0);  
tft.setRotation(2);
```



There is currently no option to mirror.

The driver chip shows the 8 possible MADCTR (MV/MX/MY) combinations.

It should be noted that the actual display orientation will also depends on the driver/display combination.

Display Data Direction	MADCTR Parameter			Image in the Host (MPU)	Image in the Driver (DDRAM)
	MV	MX	MY		
Normal	0	0	0		
Y-Mirror	0	0	1		
X-Mirror	0	1	0		
X-Mirror Y-Mirror	0	1	1		
X-Y Exchange	1	0	0		
X-Y Exchange Y-Mirror	1	0	1		
X-Y Exchange X-Mirror	1	1	0		
X-Y Exchange X-Mirror Y-Mirror	1	1	1		

The setting that works for the LilyGo_T_Display_S3 display is `tft.setRotation(3);`

This writes the following data to the display driver.

```
writedata(TFT_MAD_MV | TFT_MAD_MY | TFT_MAD_COLOR_ORDER);
```

How to support different displays?

A possible way forward to allow a single code base to be used for different displays, assuming the driver chips are bitmap compatible.

- Update the TFT_eSPI library file ST7789_Rotation.h to allow all 8 rotation options.

For example, a hack of the file to mirror the display that I have.

```
//writedata(TFT_MAD_MV | TFT_MAD_MY | TFT_MAD_COLOR_ORDER);  
writedata(TFT_MAD_MV | TFT_MAD_COLOR_ORDER);
```



- In the main code adjust the setting to work correctly for the display. For the receiver that I have the settings required are shown below.

```
// TFT display setup  
tft.begin();  
tft.invertDisplay(1); // Options 0, 1  
tft.setRotation(3); // Options 0 to 7
```

- It may also be possible to update the code to allow the settings to be modified on the fly. For example, the USB serial port could be used to control this.

Command "1": Toggle the tft.invertDisplay(*) between 0 and 1
Command "2": Step through tft.setRotation(*) between 0 and 7
Command "3": Save settings

In some cases it might be quite handy to have a rotated/mirrored option.

Notes

It should also be possible to write to the MADCTL register directly, but other parameters may need to be accounted for.

It may also be necessary to provision for setting the RGB order for different displays.